








Time Series Analysis

Chapter 7: Cointegration and VECM



Daniel Traian PELE

Bucharest University of Economic Studies
IDA Institute Digital Assets
Blockchain Research Center
AI4EFin Artificial Intelligence for Energy Finance
Romanian Academy, Institute for Economic Forecasting
MSCA Digital Finance

Acronyms used in this material

- **ACF**: Autocorrelation Function
- **ADF**: Augmented Dickey–Fuller (test)
- **AI**: Artificial Intelligence
- **AIC**: Akaike Information Criterion
- **AR**: AutoRegressive (model)
- **ARCH**: AutoRegressive Conditional Heteroskedasticity
- **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- **BIC**: Bayesian Information Criterion
- **BNR**: Banca Națională a României (the National Bank of Romania)
- **BVB**: Bursa de Valori București (the Bucharest Stock Exchange)
- **DHSY**: Davidson, Hendry, Srba and Yeo (1978)
- **DOLS**: Dynamic Ordinary Least Squares
- **DW**: Durbin–Watson (statistic)
- **ECM**: Error Correction Model
- **FRED**: Federal Reserve Economic Data
- **GDP**: Gross Domestic Product
- **HICP**: Harmonised Index of Consumer Prices
- **LR**: Likelihood Ratio (test)
- **OLS**: Ordinary Least Squares
- **PPP**: Purchasing Power Parity
- **RMSE**: Root Mean Squared Error
- **ROBOR**: Romanian Interbank Offer Rate
- **SE**: Standard Error
- **UK**: United Kingdom
- **US**: United States
- **VAR**: Vector AutoRegression
- **VECM**: Vector Error Correction Model

Today's question and route

- **Question**: two prices both wander like random walks; can a combination of them be stable, and how do we model it?
  - if yes, the series share a common stochastic trend and a long-run equilibrium: regressions in levels are meaningful, not spurious
- **Route** of the chapter
  - the idea of cointegration; the Engle–Granger two-step method with the right critical values; error correction models
  - the VECM and the Granger representation theorem; the Johansen tests; estimating and reading β and α; forecasting
  - applications: purchasing power parity, interest rates, three Central European currencies, pairs trading on the Bucharest Stock Exchange
- We build on Chapter 3 (unit roots, spurious regression) and Chapter 6 (VAR models)

Learning outcomes

- Define cointegration and explain it as a common stochastic trend
- Run the Engle–Granger and Phillips–Ouliaris tests and compare them with the right (not the Dickey–Fuller) critical values
- Estimate an error correction model and interpret the speed of adjustment and the half-life
- Write a VAR as a VECM, state the Granger representation theorem and choose the cointegration rank with the Johansen tests
- Interpret cointegrating vectors and adjustment coefficients, and decide when a VECM forecasts better than a VAR in differences
- Judge an application (parity conditions, pairs trading) out of sample and net of costs

Reading and tools

- Textbook: Huang and Petukhina (2022), Ch. 9 (nonstationarity and cointegration)
  - theory: Hamilton (1994), Ch. 19–20; Lütkepohl (2005), Ch. 6–8; the likelihood approach in Johansen (1995) and Juselius (2006)
- Landmark papers: Engle and Granger (1987), Johansen (1988), Johansen (1991); Granger's Nobel lecture Granger (2004)
- Python Quantlets of this chapter: Quantlets/Ch_07
  - `coint`, `coint_johansen`, `VECM` and `VAR` from `statsmodels`; `phillips_ouliaris` from `arch`
- Lecture notebook: open in Google Colab
- Video course: Applied Time Series Analysis with Python

From spurious regression to cointegration

- Chapter 3: regressing one $I(1)$ series on another, **unrelated** $I(1)$ series gives large t, high R^2 and a Durbin–Watson statistic near 0 Granger and Newbold (1974)
  - the error $y_t - a - bx_t$ is itself $I(1)$ for every b: nothing ties the two series together
- The exception: for one value of b, the combination $u_t = y_t - a - bx_t$ is **stationary**
  - then y_t and x_t are **cointegrated** Granger (1981), Engle and Granger (1987): the regression in levels estimates a real long-run relation
  - the two series may drift apart for a while, but u_t always pulls them back: an **equilibrium**
- Today: how to recognise this case, how to test it, and how to model it

A drunk and her dog

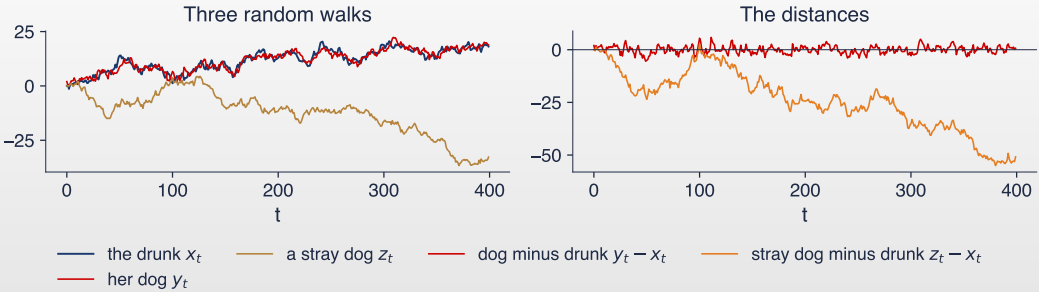


A walk with a dog: two paths that never drift far apart

Photo: Amin (2018); CC BY-SA 4.0; Wikimedia Commons

- Murray (1994): a drunk leaves the pub and wanders: her path x_t is a random walk
  - her dog also wanders: y_t is a random walk too
  - the distance between two **independent** walkers grows without limit, like $\sqrt{t}$
- But the dog hears her call and runs back; she hears the dog and turns towards it
  - each corrects part of the gap: $\Delta x_t = c\,(y_{t-1} - x_{t-1}) + u_t$, $\Delta y_t = d\,(x_{t-1} - y_{t-1}) + w_t$; $c, d > 0$: the shares of the gap closed per step; u_t, w_t: independent shocks
  - both paths remain unpredictable, but the distance $y_t - x_t$ stays bounded: **cointegration** with **error correction**

The drunk and her dog by simulation



- $T = 400$, Gaussian shocks; $c = 0.15$, $d = 0.25$; a stray dog z_t is an independent random walk with the same shocks' variance

TSA_ch7_common_trends

Interpreting the simulation: the distances

- Each path is $I(1)$: ADF on x_t gives p $= 0.21$
  - the distance $y_t - x_t$ is an AR(1) with $\rho = 1 - c - d = 0.6$: ADF p < 0.001, standard deviation 1.9
  - the distance to the stray dog wanders: ADF p $= 0.81$, standard deviation 14.1
- Half of a gap disappears in $\ln 0.5 / \ln 0.6 = 1.4$ steps: the **half-life** (Section 3)
  - knowing where the drunk is tells us where to look for the dog, even though neither path can be forecast

Definition of cointegration

- **Definition** Engle and Granger (1987): the components of $\mathbf{y}_t = (y_{1t}, \ldots, y_{nt})^\top$ are **cointegrated of order (1,1)**, $\mathbf{y}_t \sim CI(1,1)$, if
  - every component is $I(1)$ (Chapter 3: stationary after one difference), and
  - some linear combination $\beta^\top \mathbf{y}_t$, with $\beta \neq 0$, is $I(0)$: stationary
- β is the **cointegrating vector**; $u_t = \beta^\top \mathbf{y}_t - \mu$ is the **equilibrium error** (μ: its mean; $\top$: transpose)
  - β is unique only up to scale: 2β works as well; we **normalise** one coefficient to 1, e.g. $\beta = (1, -b)^\top$
  - with n variables there can be up to $r = n - 1$ independent cointegrating vectors; r is the **cointegration rank**
- Cointegration is a property of the **levels**: differencing the data destroys the information in $\beta^\top \mathbf{y}_t$

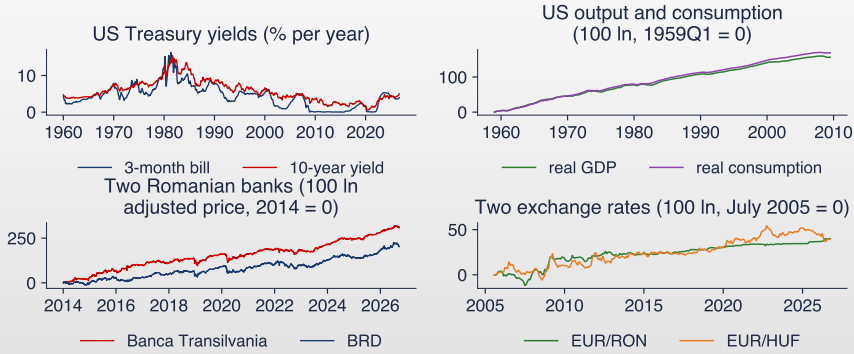
Common stochastic trends

- Write $x_t = x_{t-1} + \varepsilon_t$ (a random walk) and $y_t = b\,x_t + u_t$, with u_t stationary
  - both series contain the same stochastic trend $\sum_{s \le t} \varepsilon_s$; $y_t - b x_t = u_t$ removes it
- **Common trends** representation Stock and Watson (1988): n variables with r cointegrating vectors are driven by $n - r$ common stochastic trends
  - $r = 0$: n separate trends, no cointegration; $r = n - 1$: one trend shared by all
  - example: three yields (1, 5 and 10 years) with $r = 2$ share one trend, the **level** of interest rates (Section 5)
- **Worked example**: x_t a random walk, $y_t = 2x_t + u_t$, $z_t = -x_t + v_t$ (u_t, v_t stationary)
  - $n = 3$, one common trend, so $r = 2$: in the order (x_t, y_t, z_t), e.g. $\beta_1 = (-2, 1, 0)^\top$ and $\beta_2 = (0, 1, 2)^\top$ (since $y_t - 2x_t = u_t$ and $y_t + 2z_t = u_t + 2v_t$)

Where economics predicts cointegration

- **Purchasing power parity**: $s_t = p_t - p_t^* + q_t$, with the real exchange rate q_t stationary (Section 8)
  - s_t: log exchange rate (lei per euro); p_t, p_t^*: log price levels at home and abroad
- **Term structure of interest rates**: by the expectations hypothesis, the spread between a long and a short yield is stationary Campbell and Shiller (1987)
- **Consumption and income**: the "great ratio" $c_t - y_t$ (logs) is stable if consumption follows permanent income
- **Spot and futures prices**, an index and a fund that tracks it: arbitrage keeps the gap small
- **Shares of similar firms**: the same sector shocks; the basis of pairs trading (Section 9)

Four pairs of trending series



- US Treasury yields (FRED, monthly, since 1960); US real GDP and consumption (statsmodels macrodata, 1959–2009)
- Banca Transilvania and BRD (adjusted daily prices); EUR/RON and EUR/HUF (BNR reference rates, month-end)

TSA_ch7_common_trends

Interpreting the four pairs

- Each series is $I(1)$ (Chapter 3); the question is whether the two series of a pair move **together** in the long run
  - Engle–Granger test (Section 2), 5% critical value about -3.34
- Yields: $\tau = -3.35$, p $= 0.048$: borderline; consumption and income: $\tau = -3.54$, p $= 0.029$
- The two banks: $\tau = -4.33$, p $= 0.002$: cointegrated in this sample
- EUR/RON and EUR/HUF: $\tau = -2.47$, p $= 0.294$: **no evidence** of cointegration
  - although they rose by similar amounts and $R^2 = 0.79$ in levels: a common direction is not a common trend

Question for the room

- **Question**: two $I(1)$ series have a correlation of 0.95 in levels. Are they cointegrated?
  - **Answer**: we cannot tell: unrelated random walks often have high correlations (Chapter 3); only a test on the equilibrium error decides
- **Question**: two cointegrated series have a correlation of 0.2 between their daily changes. Is this a contradiction?
  - **Answer**: no: cointegration is about the long run; in the short run the changes can be almost unrelated

Recap: The idea of cointegration

- $I(1)$ series are cointegrated if a combination $\beta^\top \mathbf{y}_t$ is stationary: a long-run equilibrium
- Equivalently, they share common stochastic trends: n variables, r vectors, $n - r$ trends
- Theory suggests cointegration (parity conditions, spreads, great ratios), but the data must confirm it

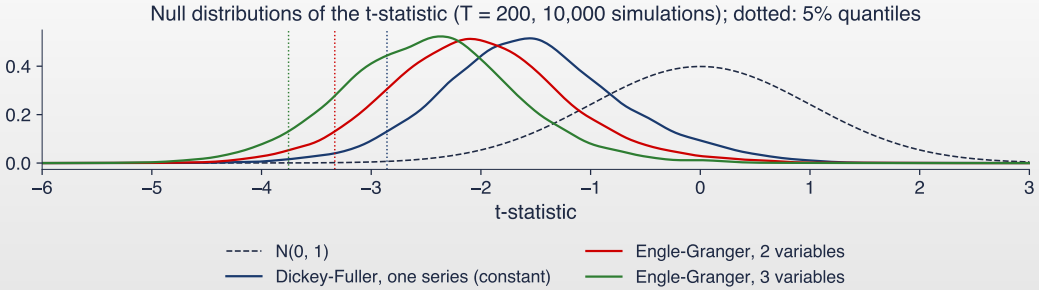
Two steps

- **Step 1**: estimate the long-run relation by OLS (ordinary least squares) in levels: $y_t = a + b\,x_t + u_t$
  - keep the residuals $\hat{u}_t = y_t - \hat{a} - \hat{b}x_t$
  - if the series are cointegrated, $\hat{b}$ is **super-consistent**: its error shrinks like $1/T$, faster than the usual $1/\sqrt{T}$
- **Step 2**: an ADF test (Chapter 3) on the residuals, without a constant: $\Delta\hat{u}_t = \gamma\,\hat{u}_{t-1} + \sum_{j=1}^{k}\delta_j\Delta\hat{u}_{t-j} + e_t$
  - H_0: $\gamma = 0$, the residuals have a unit root: **no cointegration**; H_1: $\gamma < 0$: cointegration
  - δ_j: coefficients of the k lagged differences, which absorb the autocorrelation of $\hat{u}_t$; e_t: white noise
  - statistic $\tau = \hat{\gamma}/\mathrm{SE}(\hat{\gamma})$, SE: standard error; reject H_0 when τ is below the critical value
- Pretest: each series must be $I(1)$; cointegration between an $I(0)$ and an $I(1)$ series is impossible

The reason the Dickey–Fuller table is wrong here

- OLS chooses $\hat{b}$ to make the residual variance as small as possible
  - so $\hat{u}_t$ looks more stationary than the true u_t, even when there is no cointegration
  - under H_0 the statistic τ is shifted to the left of the Dickey–Fuller distribution
- The critical values depend on the number of variables n and on the deterministic terms of step 1
  - tables: Engle and Granger (1987) (by simulation), then MacKinnon (1994) and MacKinnon (2010), used by `statsmodels` (`coint`)
- With β **known** from theory (e.g. a spread, $\beta = (1, -1)^\top$), nothing is estimated: the plain ADF test and its table apply

The null distribution by simulation



- 10 000 samples of $T = 200$ independent Gaussian random walks; step 1 with a constant, step 2 without lags

TSA_ch7_engle_granger

Interpreting the simulation: the critical values

- 5% quantiles: -2.86 (one series, Dickey–Fuller), -3.33 (2 variables), -3.75 (3 variables)
  - MacKinnon's values for $T = 200$: -2.88; -3.37; -3.78: the simulation reproduces them
- Using the Dickey–Fuller value -2.88 for the residuals would reject a true H_0 in 14.5% of samples with 2 variables and 30.3% with 3
  - instead of 5%: we would "find" cointegration far too often
- More variables give OLS more freedom to fit, hence more negative critical values

Critical values of the Engle–Granger test

Variables n	constant			constant and trend		
	1%	5%	10%	1%	5%	10%
1 (Dickey–Fuller)	−3.43	−2.86	−2.57	−3.96	−3.41	−3.13
2	−3.90	−3.34	−3.04	−4.33	−3.78	−3.50
3	−4.29	−3.74	−3.45	−4.66	−4.12	−3.84
4	−4.64	−4.10	−3.81	−4.97	−4.43	−4.15

- Asymptotic values ($T \to \infty$) of MacKinnon (2010); for finite T the values are slightly more negative
- Read the row of the **total** number of variables in the regression, y_t included
- Constant and trend in step 1: when the series have drifts that the relation does not cancel

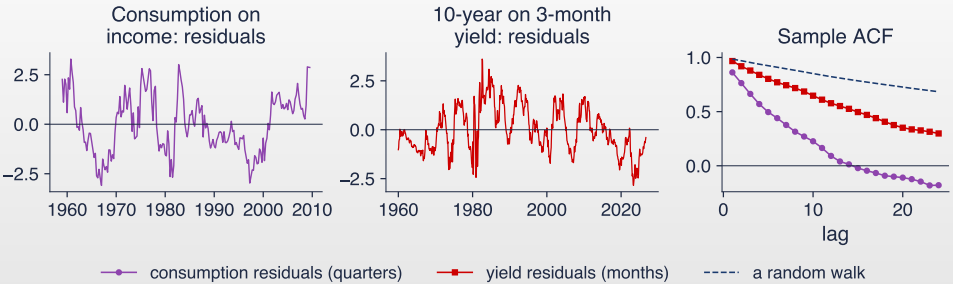
The Phillips–Ouliaris test

- Phillips and Ouliaris (1990): the same step 1 and the same H_0 (no cointegration)
  - step 2: a Dickey–Fuller regression **without** lagged differences; the t-statistic is corrected with the long-run variance of the residuals (as in the Phillips–Perron test of Chapter 3)
  - two versions: Z_t (corrected t-ratio) and Z_α (corrected $T\hat{\gamma}$); the critical values of Z_t are those of Engle–Granger
- Reasons for two tests
  - Engle–Granger depends on the number of lags k chosen by AIC; Phillips–Ouliaris depends on the bandwidth of the long-run variance
  - when they agree, the verdict is robust; when they disagree, look at the residual plot and at the sample
- In Python: `coint(y, x)` from `statsmodels`; `phillips_ouliaris(y, x, test_type="Zt")` from `arch`

Worked example: US consumption and income

- Data: c_t, $y_t = 100 \times$ log of real consumption and real GDP, quarterly, 1959Q1–2009Q3 (203 quarters)
  - pretest: ADF with trend, p = 0.31 for c_t and 0.39 for y_t: both $I(1)$
- Step 1: $\hat{c}_t = -107.6 + 1.075\, y_t$, $R^2 = 0.9992$, DW $= 0.24$
  - DW: the Durbin–Watson statistic of the residuals, about 2 without autocorrelation and close to 0 for very persistent residuals
  - the slope is close to 1: a stable consumption share; but R^2 and DW alone prove nothing
- Step 2: $\tau = -3.54$ with 0 lags; 5% critical value for $n = 2$, $T = 203$: -3.37
  - $-3.54 < -3.37$: reject "no cointegration" at 5% (p = 0.029); Phillips–Ouliaris: $Z_t = -3.52$, p = 0.031
  - with the Dickey–Fuller value (-2.88) the decision would be the same here, but for the wrong reason

The step-1 residuals



- Left and centre: $\hat{u}_t$ of consumption on income and of the 10-year on the 3-month Treasury yield; right: their sample ACF and that of a simulated random walk

TSA_ch7_engle_granger

Interpreting the residuals

- Consumption: $\hat{\rho}(1) = 0.86$ but $\hat{\rho}(8) = 0.31$: deviations of 1–3% that shrink markedly within about two years
- Yields: $\hat{\rho}(1) = 0.97$, $\hat{\rho}(12) = 0.58$: very persistent deviations; a borderline test ($\tau = -3.35$, p = 0.048)
  - high persistence is the usual situation: the power of the test against ρ close to 1 is low (Chapter 3)
- A random walk decays much more slowly: the ACF alone cannot settle the question; the test can

Engle–Granger and Phillips–Ouliaris on real data

Regression	Freq.	T	$\hat{b}$	τ (k)	p	Z_t	p
US consumption on income	quarterly	203	1.07	−3.54 (0)	0.029	−3.52	0.031
US: 10-year on 3-month yield	monthly	801	0.86	−3.35 (20)	0.048	−4.04	0.006
TLV on BRD (since 2014)	daily	3 176	1.36	−4.33 (3)	0.002	−4.14	0.005
BRD on TLV (since 2014)	daily	3 176	0.71	−4.16 (3)	0.004	−4.02	0.007
TLV on BRD (since 2010)	daily	4 187	1.68	−1.82 (3)	0.622	−1.83	0.619
EUR/RON on EUR/HUF	monthly	255	0.71	−2.47 (0)	0.294	−2.43	0.309
Romania: consumption on GDP	quarterly	126	1.46	−2.32 (8)	0.363	−6.80	< 0.001

- τ: Engle–Granger statistic (k lags by AIC); p-values of MacKinnon (2010) for 2 variables; Z_t: Phillips–Ouliaris
- Banks: adjusted prices from data/market; Romania: household consumption and GDP, Eurostat, chain-linked volumes, seasonally adjusted, since 1995

Interpreting the table

- **Normalisation**: TLV on BRD and BRD on TLV give different τ (-4.33 and -4.16) and slopes that are not exact inverses
  - Engle–Granger depends on which variable is on the left; here the verdict is the same
- **The sample**: since 2010 the test finds no cointegration for the same banks ($p = 0.622$); since 2014 it does ($p = 0.002$)
  - choosing the start date after looking at the data is a form of data snooping
- **The two tests disagree** for Romania: Engle–Granger $p = 0.363$ with 8 lags, Phillips–Ouliaris $p < 0.001$
  - the slope 1.46 is far from 1: the consumption share rose strongly; Seminar 7 (B2) discusses this case

Limits of the Engle–Granger method

- **At most one** cointegrating vector: with $n \geq 3$ variables there may be several (Sections 4–5)
- **Normalisation**: the results depend on the variable put on the left
- **Inference on** $\hat{b}$: super-consistent, but biased in small samples, and its OLS t-statistics are **not** valid
  - remedy: dynamic OLS (DOLS) Stock and Watson (1993), adding leads and lags of Δx_t, gives valid standard errors
- **Two-step errors**: mistakes in step 1 carry over to step 2; the power is low against slow adjustment
  - the Johansen method (Section 5) estimates everything at once, by maximum likelihood

Case study: Engle and Granger (1987)



Robert Engle (2022); Nobel Prize in Economics 2003

Photo: Jon Demske (2022); CC BY-SA 4.0; Wikimedia Commons

- **Question**: how to model $I(1)$ series that are tied in the long run, without spurious regressions and without differencing away the long run?
  - Engle and Granger (1987), written at the University of California, San Diego, one of the most cited papers in econometrics
- **Contribution**
  - the definition of cointegration and the representation theorem (Section 4): cointegration $\Leftrightarrow$ error correction
  - the two-step estimator and residual-based tests with simulated critical values
  - applications to US series such as consumption and income, prices and wages, short and long interest rates

Recap: The Engle–Granger method

- Step 1: OLS in levels (super-consistent $\hat{b}$); step 2: ADF on the residuals, H_0: no cointegration
- Critical values of Engle–Granger (MacKinnon), more negative than Dickey–Fuller and dependent on n
- Check with Phillips–Ouliaris, with the reverse normalisation and with another sample

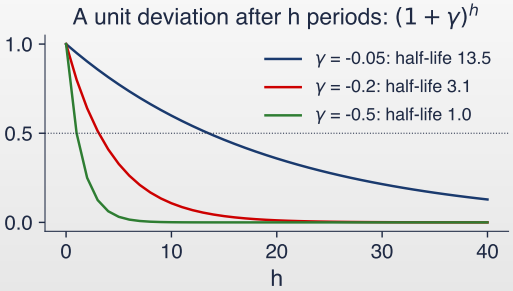
The error correction model

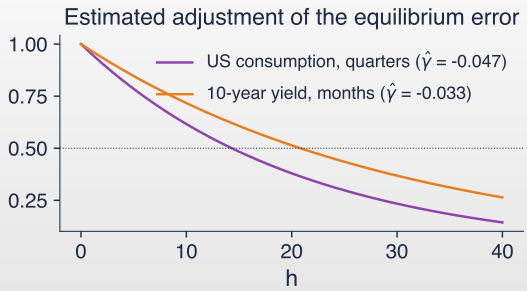
- With $u_{t-1} = y_{t-1} - a - bx_{t-1}$ (last period's equilibrium error), the **error correction model** (ECM) is
  - $\Delta y_t = c + \gamma\, u_{t-1} + \delta_0\, \Delta x_t + \sum_{j\geq 1}(\phi_j \Delta y_{t-j} + \delta_j \Delta x_{t-j}) + \varepsilon_t$
- Every term is stationary: OLS and the usual t-tests are valid (with $\hat{u}_{t-1}$ from step 1)
  - b: the **long-run** effect of x on y; δ_0: the **short-run** (same-period) effect
  - γ: the **speed of adjustment**; error correction requires $-2 < \gamma < 0$
  - c: constant; ϕ_j, δ_j: short-run coefficients of the lagged differences; ε_t: white noise
- Reading: if y_{t-1} was above equilibrium ($u_{t-1} > 0$) and $\gamma < 0$, then y_t tends to fall: a share $|\gamma|$ of the gap is closed each period

Speed of adjustment and half-life

- Without new shocks, the equilibrium error evolves as $u_t = (1+\gamma)\, u_{t-1}$, so $u_h = (1+\gamma)^h u_0$
  - (in a bivariate system where only y adjusts)
- **Half-life**: the number of periods after which half of a deviation is gone
  - $(1+\gamma)^h = 0.5 \;\Rightarrow\; h_{1/2} = \ln 0.5 / \ln(1+\gamma)$
- **Worked example**: $\gamma = -0.2$, $h_{1/2} = \ln 0.5 / \ln 0.8 = 3.1$ periods
  - in months: about three months for half of a deviation; in quarters: three quarters
- Compare half-lives only in the same time unit: a monthly and a quarterly γ are not comparable

Half-lives: theory and estimates





⊡ Left: $(1+\gamma)^h$ for three speeds; right: the decay implied by the estimated ECMs of US consumption (quarters) and of the 10-year yield (months)

TSA_ch7_error_correction

Interpreting the estimated ECMs

- Consumption: $\hat{\gamma} = -0.047$ ($t = -1.75$), half-life 14.3 quarters; $\hat{\delta}_0 = 0.54$
  - half of a rise in income reaches consumption in the same quarter; the rest comes slowly, and the adjustment term is only marginally significant
- 10-year yield: $\hat{\gamma} = -0.033$ ($t = -4.94$), half-life 20.8 months; $\hat{\delta}_0 = 0.38$
  - the 3-month rate also reacts to the same gap: $\hat{\gamma} = 0.024$ ($t = 2.08$), with the opposite sign: both rates move towards each other
- When both variables adjust, a single-equation ECM tells only half of the story: hence the VECM (Section 4)

Worked example: one quarter of consumption

- Estimated ECM (main terms): $\Delta c_t = 0.418 + 0.54\,\Delta y_t - 0.047\,\hat{u}_{t-1} + \ldots$, with $\hat{u}_{t-1} = c_{t-1} - 1.075\,y_{t-1} - \hat{a}$
- Scenario: consumption is 2% above its equilibrium ($\hat{u}_{t-1} = 2$) and income grows by 1% this quarter
  - short-run effect: $0.537 \times 1 = 0.537$; error correction: $-0.047 \times 2 = -0.095$
  - predicted growth of consumption: about 0.86% (the lagged differences are set to 0)
- Without the error correction term, the model would ignore that consumption is already too high

Case study: the consumption function of DHSY (1978)

- **Question**: why did the consumption functions of the 1970s, estimated in levels or in differences, forecast UK consumers' expenditure so badly?
  - Davidson et al. (1978) (Davidson, Hendry, Srba and Yeo, "DHSY"): quarterly UK data on expenditure and income
- **Method**: a model in differences plus the lagged log ratio of expenditure to income, the error correction term
  - a general-to-specific search, starting from a large dynamic model
- **Legacy**: the ECM form came first; Granger (1981) and Engle and Granger (1987) then showed why it works: the two series are cointegrated
  - a test of cointegration can also use the t-statistic of γ in the ECM, with its own critical values Banerjee, Dolado and Mestre (1998)

Recap: Error correction models

- ECM: Δy_t on Δx_t, lagged differences and $\hat{u}_{t-1}$; all terms stationary
- $\gamma < 0$: speed of adjustment; half-life $\ln 0.5 / \ln(1+\gamma)$, in the units of the data
- Short run (δ_0) and long run (b) in one equation; if several variables adjust, we need a system

From a VAR to a VECM

- Chapter 6: a VAR(p) in levels, $\mathbf{y}_t = \mathbf{c} + A_1\mathbf{y}_{t-1} + \cdots + A_p\mathbf{y}_{t-p} + \mathbf{u}_t$
  - subtract $\mathbf{y}_{t-1}$ and regroup the lags: the same model in **vector error correction** form (VECM)
- $\Delta\mathbf{y}_t = \mathbf{c} + \Pi\,\mathbf{y}_{t-1} + \sum_{i=1}^{p-1}\Gamma_i\,\Delta\mathbf{y}_{t-i} + \mathbf{u}_t, \quad \Pi = A_1 + \cdots + A_p - I, \quad \Gamma_i = -(A_{i+1} + \cdots + A_p)$
  - Π: the long-run matrix (it multiplies the levels); Γ_i: the short-run matrices; I: the identity matrix
- **Derivation** for $p = 2$: $\mathbf{y}_t - \mathbf{y}_{t-1} = (A_1 - I)\mathbf{y}_{t-1} + A_2\mathbf{y}_{t-2} + \mathbf{u}_t$; add and subtract $A_2\mathbf{y}_{t-1}$:
  - $\Delta\mathbf{y}_t = (A_1 + A_2 - I)\,\mathbf{y}_{t-1} - A_2\,\Delta\mathbf{y}_{t-1} + \mathbf{u}_t$: $\Pi = A_1 + A_2 - I$, $\Gamma_1 = -A_2$
- The VECM is a rewriting, not a new model: all the information about the long run sits in the matrix Π

The rank of Π

- $\Delta \mathbf{y}_t$ is stationary; so $\Pi \mathbf{y}_{t-1}$ must be stationary too. Three cases, by $r = \text{rank}(\Pi)$:
- $r = 0$: $\Pi = 0$; no cointegration; the right model is a **VAR in differences**
- $r = n$: Π has full rank; $\mathbf{y}_t$ is already stationary; a **VAR in levels** (Chapter 6)
- $0 < r < n$: **cointegration** with r vectors; $\Pi = \alpha\beta^\top$ with α, β of size $n \times r$
  - $\beta^\top \mathbf{y}_{t-1}$: the r equilibrium errors; α: the **adjustment coefficients** (loadings), how strongly each variable reacts to each error
  - only the product is unique: $\alpha\beta^\top = (\alpha Q)(\beta Q^{-\top})^\top$ for any invertible Q; we normalise β
- Testing for cointegration = testing the rank of Π (Section 5)

The Granger representation theorem

- **Theorem** Engle and Granger (1987): if $\mathbf{y}_t$ is $I(1)$ and cointegrated with rank r, then it has a VECM representation with $\Pi = \alpha\beta^\top$, and conversely
  - cointegration $\Leftrightarrow$ error correction: the equilibrium error must feed back into the changes
- **Consequences**
  - a VAR in differences omits $\alpha\beta^\top \mathbf{y}_{t-1}$: it is **misspecified** when the series are cointegrated
  - $\alpha \neq 0$: at least one variable adjusts, so there is Granger causality (Chapter 6) in at least one direction
  - the levels are driven by $n - r$ common stochastic trends; the effects of shocks on the levels do not die out
- A VAR in levels is not wrong either: it is consistent, but it ignores the restriction $\text{rank}(\Pi) = r$ and its tests are non-standard

2003: the Nobel Prize for cointegration



Clive Granger (1934–2009)

Photo: Olaf Storbeck (2008); CC BY-SA 2.0; Wikimedia Commons



The Sir Clive Granger Building, University of Nottingham

Photo: mattbuck (2012); CC BY-SA 3.0; Wikimedia Commons

- The 2003 prize: Granger for cointegration, shared with Robert Engle for ARCH (Chapter 5)
- Granger studied and began his career at Nottingham
- Nobel lecture: Granger (2004): "time series analysis, cointegration, and applications"

Worked example: a bivariate VECM

- $\Delta y_{1t} = -0.10\,(y_{1,t-1} - y_{2,t-1}) + u_{1t}, \quad \Delta y_{2t} = 0.10\,(y_{1,t-1} - y_{2,t-1}) + u_{2t}$
  - $\beta = (1,-1)^\top$, $\alpha = (-0.10, 0.10)^\top$; $\Pi = \alpha\beta^\top = \begin{pmatrix} -0.10 & 0.10 \\ 0.10 & -0.10 \end{pmatrix}$, $\det \Pi = 0$, rank 1
- If y_1 is above y_2: y_1 falls ($\alpha_1 < 0$) and y_2 rises ($\alpha_2 > 0$): both close the gap
- The gap $z_t = y_{1t} - y_{2t}$: $z_t = (1 + \beta^\top\alpha)\, z_{t-1} + (u_{1t} - u_{2t}) = 0.8\, z_{t-1} + \ldots$
  - a stationary AR(1); half-life $\ln 0.5 / \ln 0.8 = 3.1$ periods; the condition is $-2 < \beta^\top\alpha < 0$
- The average $(y_{1t} + y_{2t})/2$ changes by $(u_{1t} + u_{2t})/2$: a random walk, the common trend

Deterministic terms: five cases

Case	In the VECM	Implication for the data	`statsmodels`
1	no constant, no trend	no drift; relations through the origin	`n`
2	constant only inside $\beta^\top \mathbf{y}_{t-1}$	no drift; equilibrium errors with a non-zero mean (yields, spreads)	`ci`
3	unrestricted constant	linear trends in the levels, none in the relations (GDP, prices)	`co`
4	constant, and a trend inside $\beta^\top \mathbf{y}_{t-1}$	trends in the levels and in the relations	`coli`
5	unrestricted constant and trend	quadratic trends in the levels; rarely plausible	`colo`

- The critical values of the Johansen tests change with the case Johansen (1995); choose the case from the plot and from theory, before testing
- `coint_johansen` of `statsmodels` implements case 1 (`det_order=-1`) and case 3 (`det_order=0`), the case used in this chapter

Weak exogeneity

- If a row of α is zero, $\alpha_i = 0$, the variable y_i does **not** react to the equilibrium errors
  - y_i is **weakly exogenous** for β: the other variables do all the adjusting
  - y_i pushes the common trend; e.g. Euribor for Romanian interest rates (Seminar 7, B4)
- Test: a t-test on α_i (one cointegrating vector) or a likelihood-ratio (LR) test of $\alpha_i = 0$ Johansen (1991)
  - the statistic is asymptotically χ^2 with r degrees of freedom, once r is fixed
- Use: if x_t is weakly exogenous, a single-equation ECM for y_t (Section 3) loses no information about β

Recap: The VECM

- $\Delta \mathbf{y}_t = \alpha \beta^\top \mathbf{y}_{t-1} + \sum \Gamma_i \Delta \mathbf{y}_{t-i} + \mathbf{u}_t$: a VAR rewritten; $\text{rank}(\Pi) = r$
- Granger representation: cointegration $\Leftrightarrow$ error correction; a VAR in differences omits the long run
- β: equilibria; α: who adjusts; deterministic terms: five cases; $\alpha_i = 0$: weak exogeneity

The idea: reduced-rank regression



University of Copenhagen, where Søren Johansen and Katarina Juselius developed the method

Photo: Per Meistrup (2011); CC BY-SA 4.0; Wikimedia Commons

- Johansen (1988), Johansen (1991): maximum likelihood for the VECM with Gaussian errors
  - regress $\Delta \mathbf{y}_t$ and $\mathbf{y}_{t-1}$ on the lagged differences; keep the residuals R_{0t} and R_{1t}
  - find the combinations of R_{1t} most correlated with R_{0t}: the **canonical correlations**
- Their squares are the eigenvalues $1 > \hat{\lambda}_1 \geq \cdots \geq \hat{\lambda}_n \geq 0$
  - a large $\hat{\lambda}_i$: a stationary combination that predicts the changes; $\hat{\lambda}_i \approx 0$: a combination that is still $I(1)$
  - the eigenvectors give $\hat{\beta}$; then $\hat{\alpha}$ follows by OLS

Trace and maximum-eigenvalue statistics

- **Trace test**: H_0: rank $\leq r$, against rank $= n$; $\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^{n} \ln(1 - \hat{\lambda}_i)$
  - it sums the information in the $n - r$ smallest eigenvalues: if they are all close to 0, the statistic is small and H_0 is kept
- **Maximum-eigenvalue test**: H_0: rank $= r$, against rank $= r + 1$; $\lambda_{\max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
  - it uses only the next eigenvalue, $\hat{\lambda}_{r+1}$; T: the number of observations
- **Sequential procedure**: test $r = 0$, then $r \leq 1$, ...; the rank is the first r whose H_0 is **not** rejected
  - the statistics are likelihood ratios, but their distributions are non-standard (functions of Brownian motions)
  - critical values depend on $n - r$ and on the deterministic case: Osterwald-Lenum (1992), MacKinnon, Haug and Michelis (1999)
- The two tests usually agree; if not, the trace test is preferred in most applied work

Worked example: from eigenvalues to statistics

- US yields at 1, 5 and 10 years, 801 months; effective sample $T = 801 - 2 - 1 = 798$: $\hat{\lambda} = (0.0405, 0.0229, 0.0039)$
  - $-\ln(1 - \hat{\lambda}_i)$: 0.0414; 0.0231; 0.0039
- $r = 0$: trace $= 798 \times (0.0414 + 0.0231 + 0.0039) = 54.6 > 29.80$: reject
  - max-eigenvalue: $798 \times 0.0414 = 33.0 > 21.13$: reject
- $r \leq 1$: trace $= 21.6 > 15.49$: reject; $r \leq 2$: trace $= 3.1 < 3.84$: do not reject
  - rank 2: two cointegrating vectors, one common trend
- Small eigenvalues still matter: they are multiplied by T

Johansen tests on three systems

System	T	k	$\lambda_{\text{trace}}(0)$	$\lambda_{\text{trace}}(1)$	$\lambda_{\text{trace}}(2)$	$\lambda_{\max}(0)$	$\lambda_{\max}(1)$	rank
US yields 1y, 5y, 10y	801	2	54.6	21.6	3.1	33.0	18.5	2
US output, consumption, investment	203	1	28.9	11.4	2.6	17.4	8.9	0
EUR/RON, EUR/HUF, EUR/PLN	255	1	24.2	7.9	2.7	16.3	5.2	0
5% critical value			29.80	15.49	3.84	21.13	14.26	

- Case 3 (unrestricted constant); k: lagged differences chosen by BIC on the VAR in levels; ranks by the trace test (the maximum-eigenvalue test gives the same ranks here)
- Yields: FRED, 1960–2026; output, consumption and investment: 100 ln, US 1959–2009; currencies: BNR, month-end, 2005–2026

Interpreting the three systems

- Yields: rank 2, so one common trend: the **level** of rates moves all maturities; the two spreads are stationary
  - the expectations hypothesis of the term structure Campbell and Shiller (1987) in its weak form
- Output, consumption and investment: trace 28.9 against 29.80: rank 0 at 5%, a near miss
  - balanced growth predicts rank 2 (stable shares of consumption and investment); with Engle–Granger, consumption and income were cointegrated ($p = 0.029$): low power, not proof of absence
- Currencies: trace 24.2: rank 0; three separate stochastic trends (Section 8)

The deterministic case changes the answer

Deterministic terms	$\lambda_{\text{trace}}(0)$	5%	$\lambda_{\text{trace}}(1)$	5%	$\lambda_{\text{trace}}(2)$	5%
none, case 1 (`det_order=-1`)	45.9	24.28	13.3	12.32	0.4	4.13
constant, case 3 (`det_order=0`)	54.6	29.80	21.6	15.49	3.1	3.84
constant and trend, case 5 (`det_order=1`)	60.0	35.01	23.7	18.40	4.8	3.84

- US yields at 1, 5 and 10 years: ranks 2, 2 and 3
- With a linear trend, the test says that every yield is stationary around a trend: not plausible for interest rates over 66 years
  - choose the deterministic terms from theory and from the plot, not from the result you like
- Other practical issues: the lag length k; size distortions in small samples (correction factor $(T - nk)/T$, Reinsel and Ahn (1992)); structural breaks

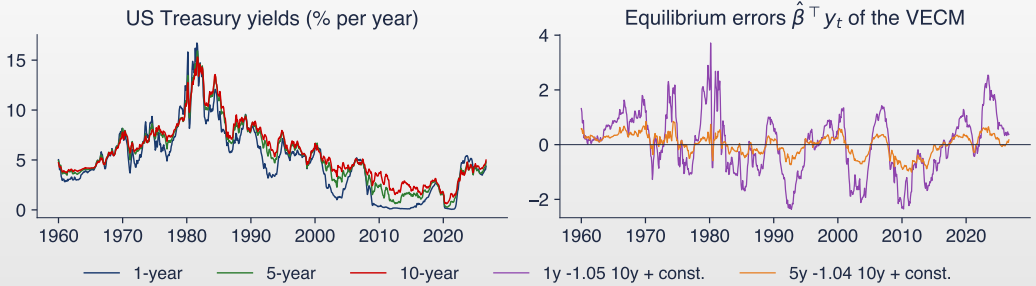
Case study: Johansen and Juselius (1990)

- **Question**: is there a stable demand for money in Denmark and Finland?
  - Johansen and Juselius (1990): real money, real income, interest rates (and inflation); quarterly data
- **Method**: the VECM by maximum likelihood; the rank by the trace and maximum-eigenvalue tests; then **tests of restrictions** on β and α
  - e.g. a unit income elasticity, $\beta = (1, -1, \cdot)$; weak exogeneity of income and interest rates
- **Legacy**: the template of the "cointegrated VAR" methodology Juselius (2006): test the rank, then test economic hypotheses as restrictions
  - one of the most cited papers in applied econometrics

Recap: The Johansen tests

- Eigenvalues of a reduced-rank regression; trace and maximum-eigenvalue statistics; sequential testing from $r = 0$
- Critical values depend on $n - r$ and on the deterministic case
- Yields: rank 2; macro aggregates: borderline; currencies: rank 0

The term-structure VECM



- US Treasury yields at 1, 5 and 10 years (FRED GS1, GS5, GS10), January 1960–September 2026; VECM with rank 2, 2 lagged differences, constant restricted to the relations (case 2)

TSA_ch7_johansen_vecm

Interpreting the cointegrating vectors

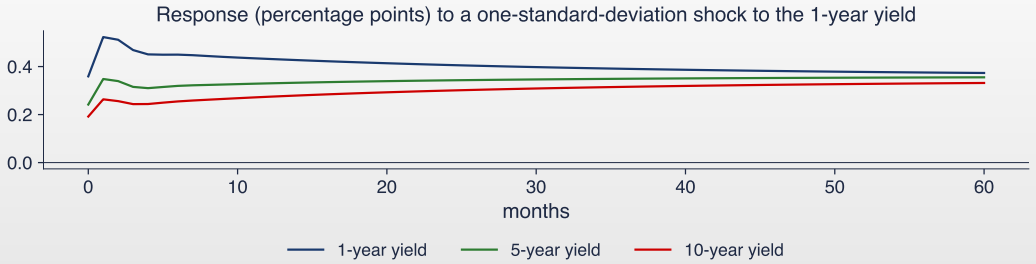
- Normalisation on the 1-year and 5-year yields: $\hat{\beta}_1^\top \mathbf{y}_t = i_t^{(1)} - 1.045\, i_t^{(10)} + 1.22$, $\hat{\beta}_2^\top \mathbf{y}_t = i_t^{(5)} - 1.044\, i_t^{(10)} + 0.59$
  - both slopes close to 1: the equilibria are almost the spreads $i^{(1)} - i^{(10)}$ and $i^{(5)} - i^{(10)}$, with mean term premia
- The equilibrium errors are stationary: ADF $p < 0.001$ and 0.008; they are large in the 1980s and around zero-rate periods
- A formal test of $\beta = (1, -1)$ for each spread is a likelihood-ratio test Johansen (1991); with super-consistent $\hat{\beta}$, values this close to 1 are informative

The adjustment coefficients

Equation	$\hat{\alpha}_{\cdot \mathbf{1}}$	t	$\hat{\alpha}_{\cdot \mathbf{2}}$	t
$\Delta i_t^{(\mathbf{1})}$ (1 year)	-0.021	-0.82	0.024	0.31
$\Delta i_t^{(\mathbf{5})}$ (5 years)	0.045	2.21	-0.122	-2.00
$\Delta i_t^{(\mathbf{10})}$ (10 years)	0.040	2.22	-0.077	-1.43

- The 1-year yield does not react to either error ($|t| < 1$): it behaves as **weakly exogenous**
  - the short end follows monetary policy; the 5- and 10-year yields do the adjusting ($|t|$ from 1.43 to 2.22)
- The equilibrium errors $\mathbf{z}_t = \hat{\beta}^\top \mathbf{y}_t$ follow $\mathbf{z}_t = (\boldsymbol{I} + \hat{\beta}^\top \hat{\alpha})\, \mathbf{z}_{t-\mathbf{1}} + \ldots$; eigenvalues 0.969 and 0.926
  - half-lives of about 22.1 and 9.0 months: slow error correction, typical of interest rates

Impulse responses of the VECM



⊡ Orthogonalised responses (Cholesky, ordering 1y, 5y, 10y; Chapter 6) to a one-standard-deviation shock to the 1-year yield

TSA_ch7_johansen_vecm

Interpreting the impulse responses

- On impact: +0.36, +0.24 and +0.19 percentage points for 1, 5 and 10 years
  - after 60 months: +0.37, +0.36 and +0.33: the effects do **not** die out and become almost equal
- A permanent shift of the common trend (the level), with stationary spreads: the signature of $n - r = 1$ common trend
- In a stationary VAR (Chapter 6) every response returns to zero; in a VECM the long-run responses are generally different from zero

Recap: β and α

- $\hat{\beta}$: normalise, read as equilibria (spreads, ratios), test economic restrictions
- $\hat{\alpha}$: who adjusts and how fast; a zero row means weak exogeneity
- Shocks have permanent effects on cointegrated levels, but not on the equilibrium errors

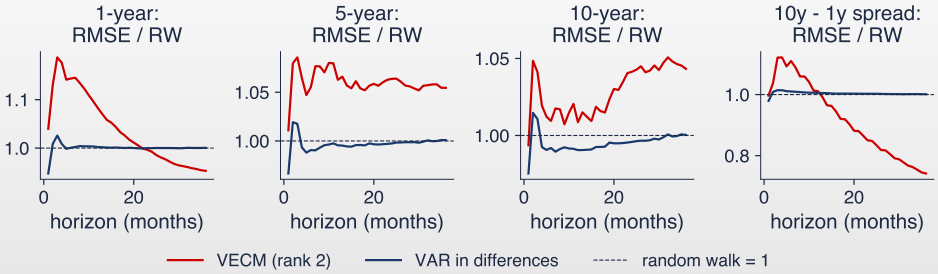
The theoretical arguments

- VECM forecasts keep the equilibria: as the horizon grows, $\hat{\beta}^\top \hat{\mathbf{y}}_{T+h}$ returns to its mean
  - a VAR in differences forgets the levels: its forecasts of the spreads stay where they are
- Engle and Yoo (1987): imposing cointegration improves long-horizon forecasts in simulated systems
- Christoffersen and Diebold (1998): the gains are concentrated in the **cointegrating combinations**
  - for the individual levels, the forecast error is dominated by the common trend, which no model can predict
  - estimation error in α, β and in the constants can make a VECM worse in practice
- Test it out of sample, with re-estimation at each forecast origin

The forecasting experiment

- Data: US yields at 1, 5 and 10 years, monthly, since 1960
- 135 forecast origins, every three months from January 1990 to July 2023; models re-estimated on the data up to each origin
  - VECM with rank 2 and a restricted constant; VAR in differences without constant, with the same lags; random walk (no change)
- Horizons of 1 to 36 months; RMSE (root mean squared error) of each model divided by that of the random walk
  - ratio below 1: better than the random walk; also for the 10-year minus 1-year spread

VECM against a VAR in differences, out of sample



- RMSE relative to the random walk by horizon; first three panels: the yields; last panel: the 10-year minus 1-year spread

TSA_ch7_forecasting

Interpreting the forecast comparison

- Levels: neither model beats the random walk clearly; the VECM is worse at short horizons (1 year at 6 months: 1.14; 5 years: 1.06)
  - the VAR in differences stays close to 1 at every horizon (1 year at 12 months: 1.00)
- Spread: the VECM gains at long horizons: 0.85 at 24 months and 0.74 at 36 months; the VAR in differences: 1.00
  - exactly the pattern of Christoffersen and Diebold (1998): cointegration helps forecast the equilibrium errors, not the trend
- Lesson: use a VECM when the object of interest is the relation (a spread, a ratio, a pair); for the levels alone, a random walk is hard to beat

Recap: Forecasting

- The VECM pulls the forecasts of the equilibrium errors back to their means; the VAR in differences does not
- Out of sample: gains for the spread at long horizons, none for the levels
- Always compare with the random walk, with re-estimation and without look-ahead

Purchasing power parity

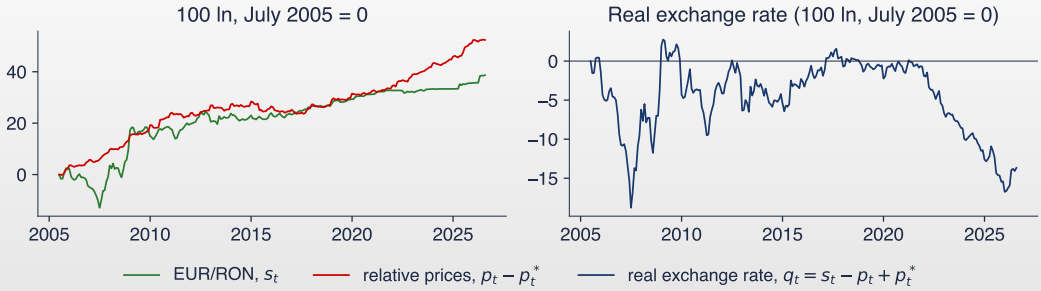


The National Bank of Romania, which publishes the reference rates

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia Commons

- **Relative PPP** (purchasing power parity): the exchange rate offsets inflation differences
  - $s_t = 100 \ln(\text{EUR/RON})$; p_t, p_t^*: $100 \ln$ of the HICP (Harmonised Index of Consumer Prices) of Romania and of the euro area
  - PPP holds in the long run if the real exchange rate $q_t = s_t - p_t + p_t^*$ is stationary: cointegration with $\beta = (1, -1, 1)$
- Reasons why it may fail
  - Balassa (1964): fast productivity growth in tradables raises prices of non-tradables in a catching-up economy: a trend real appreciation
  - even where PPP holds, deviations decay slowly; the survey Taylor and Taylor (2004) reports half-lives of several years

PPP for the leu



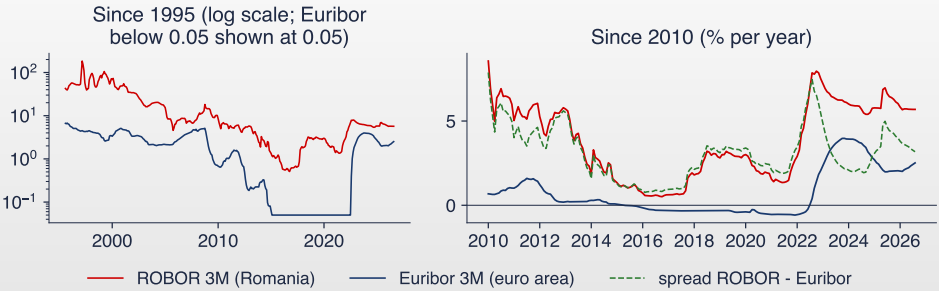
- Monthly, July 2005–August 2026: EUR/RON (monthly mean of the BNR reference rate), HICP of Romania and of the euro area (Eurostat, 2015 = 100)

TSA_ch7_parity_conditions

Interpreting PPP for the leu

- Since 2005 the leu lost 38.7 log points against the euro, while Romanian prices rose 52.4 log points more than euro-area prices
  - the real exchange rate changed by -13.7 log points: a real appreciation of the leu, concentrated after 2021
- ADF on q_t: p $= 0.24$; Engle–Granger of s_t on $p_t - p_t^*$: slope 0.89, p $= 0.34$; Johansen on (s_t, p_t, p_t^*): rank 0
- No evidence of PPP over 21 years, consistent with Balassa–Samuelson convergence; 21 years may also be too short for half-lives of several years

ROBOR and Euribor



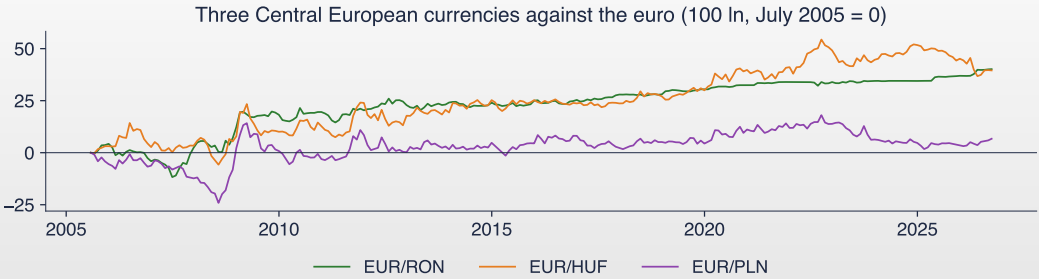
- Three-month money market rates (Eurostat), August 1995–August 2026: ROBOR 3M for Romania and Euribor 3M for the euro area

TSA_ch7_parity_conditions

Interpreting ROBOR and Euribor

- Full sample: Engle–Granger p = 0.52, Phillips–Ouliaris p = 0.010: the disinflation of the 1990s dominates and the tests disagree
- Since 2010: p = 0.047 (Engle–Granger) and 0.029 (Phillips–Ouliaris), slope 1.19, mean spread 3.11 percentage points
  - ECM for ROBOR: $\hat{\gamma} = -0.020$ ($t = -1.18$), half-life about 34 months: weak and slow adjustment
- Romanian monetary policy follows its own inflation target; the link to Euribor is loose (Seminar 7, B4, adds the Romanian 10-year yield)

Three Central European currencies



- EUR/RON, EUR/HUF and EUR/PLN, cross rates from the BNR reference rates, month-end, 100 ln

TSA_ch7_parity_conditions

Interpreting the three currencies

- Since July 2005: EUR/RON +40.1, EUR/HUF +39.5, EUR/PLN +6.7 log points; ADF does not reject a unit root at 5% for any rate (p = 0.55; 0.75; 0.08, EUR/PLN borderline at 10%)
- Johansen: trace 24.2 for $r = 0$ (critical value 29.80): no evidence of cointegration, also since 2012 (trace 12.7)
  - a common anchor (the euro) and similar shocks do not create a common trend: each central bank and each inflation path adds its own
- The right model for these rates is a VAR in differences (Chapter 6); Seminar 7 (B3) repeats the analysis

Recap: Economic examples

- No evidence of PPP for the leu over 2005–2026 (a unit root in q_t is not rejected): real appreciation of a converging economy
- ROBOR and Euribor: a loose, slow link since 2010; the sample decides the verdict
- Theory proposes β; the tests, the sample and the deterministic terms decide whether the data agree

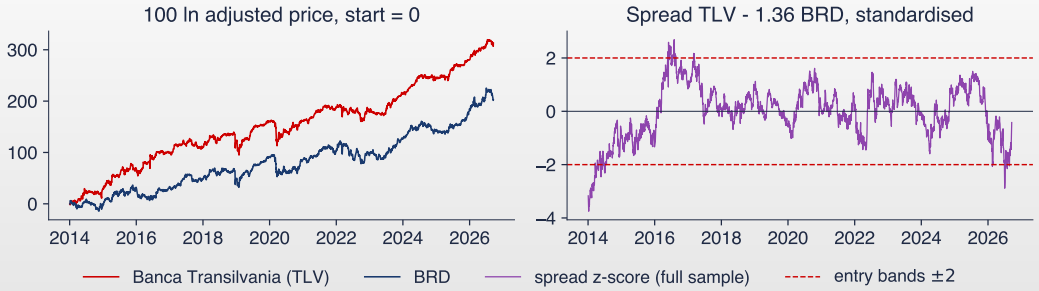
The idea of pairs trading

- Two shares exposed to the same shocks (two banks, two oil companies) may be cointegrated: their spread mean-reverts
  - when the spread is unusually wide, sell the expensive share and buy the cheap one; close when the spread is back
- A **relative-value** trade: the market direction cancels out; the profit comes from the error correction
  - Gatev, Goetzmann and Rouwenhorst (2006): the distance method on US shares, profitable over 1962–2002
  - Do and Faff (2010): the profits declined in later years
- A good test of a time-series idea: it must work **out of sample**, **after costs**, and on pairs chosen **without looking ahead**

A rule without look-ahead

- **Formation** (the previous 252 trading days): test every pair with Engle–Granger; keep the pairs with p < 0.05
  - $P_{A,t}$, $P_{B,t}$: the prices of shares A and B; r_A, r_B: their daily log returns
  - hedge ratio $\hat{b}$ by OLS of $\ln P_A$ on $\ln P_B$; mean $\bar{s}$ and standard deviation $\hat{\sigma}_s$ of the spread $s_t = \ln P_{A,t} - \hat{a} - \hat{b} \ln P_{B,t}$
- **Trading** (the next 126 days), with the formation parameters fixed: $z_t = (s_t - \bar{s})/\hat{\sigma}_s$
  - $z_t > 2$: short A, long $\hat{b}$ of B; $z_t < -2$: the opposite; close at $z_t = 0$ or at the end of the window
  - daily return per unit of gross exposure: $\pm(r_A - \hat{b}\, r_B)/(1 + |\hat{b}|)$; equal capital per selected pair
- **Costs**: 0.20% (Bucharest) and 0.05% (US) of the gross exposure traded, at each opening and closing

Banca Transilvania and BRD



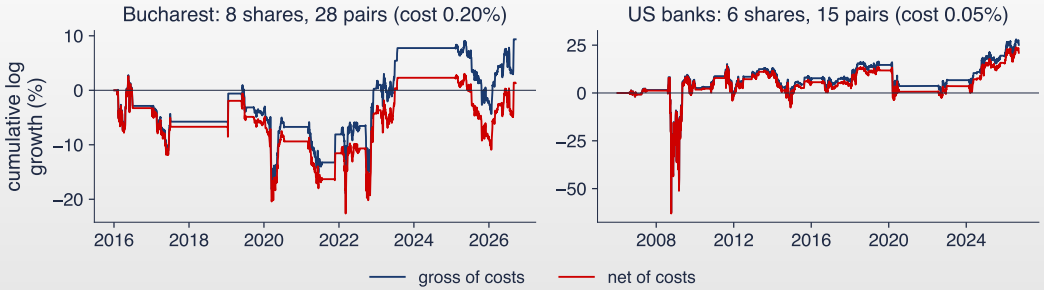
- Adjusted daily prices since 2014; right: the spread of the full-sample regression, standardised, with the entry bands

TSA_ch7_pairs_trading

Interpreting the bank pair

- Full sample: $\hat{b} = 1.36$; the spread is an AR(1) with $\hat{\rho} = 0.990$: half-life about 69 trading days
  - only 5.0% of the days lie outside ± 2: few trading opportunities, each lasting months
- Trading this picture with full-sample parameters gives 3.32% per year (Sharpe ratio 0.44), after costs
  - Sharpe ratio: the mean annual return divided by the annual standard deviation of the returns
  - an illusion: $\hat{b}$, $\bar{s}$ and $\hat{\sigma}_s$ use the future, and the pair was chosen because it looks cointegrated since 2014
- With the rolling rule, the pair passes the formation test in none of its 24 windows: one year of data is too short to detect such slow error correction

Pairs trading out of sample



- Left: 8 liquid BVB shares (TLV, BRD, SNP, TGN, TEL, EL, SNG, SNN), since 2015; right: 6 US banks (JPM, BAC, C, WFC, GS, MS), since 2005

TSA_ch7_pairs_trading

Interpreting the backtest

- Bucharest: 22 windows, pairs found in 14, 73 trades, 62% of trades profitable, but the mean return per trade is -0.26%
  - gross 1.17% per year (Sharpe 0.15); net 0.41% (Sharpe 0.05); break-even cost 0.31%
- US banks: gross 2.77% (Sharpe 0.16), net 2.57% (Sharpe 0.14), 80 trades
  - with a deep loss in the autumn of 2008: cointegration can break exactly when it matters
- Many small wins and a few large losses: the typical profile of a mean-reversion strategy

Pitfalls of pairs trading

- **Multiple testing**: with 28 pairs and a 5% test, about 1.4 pairs pass by chance in each window
- **Look-ahead**: choosing pairs, start dates or thresholds after seeing the whole sample
- **Breaks**: mergers, recapitalisations, regulation and crises change β; the spread may never come back
- **Market frictions**: short selling is limited on the BVB, borrowing shares costs a fee, bid-ask spreads are wide for small caps
- **Slow error correction**: half-lives of months mean capital tied up for months, and a long wait in a losing position

Recap: Pairs trading

- Pairs trading bets on error correction in the spread of two cointegrated shares
- In sample it looks attractive; out of sample, after costs, the profits on the BVB are close to zero
- Report the selection, the costs, the number of trades and the worst episode, not only the Sharpe ratio

Possible contribution of AI

- **Code**: a first draft of a script that runs unit-root tests, Engle–Granger, Phillips–Ouliaris and Johansen on many series and tabulates the verdicts
- **Explanation**: a second explanation of the Granger representation theorem, or of a `VECM` output table
- **Exploration**: candidate pairs on the BVB, or parity conditions for several Central European countries
- Example prompt
  - `Write Python code that loads monthly US Treasury yields GS1, GS5 and GS10 from FRED, runs the Johansen trace test with a constant and BIC lags, fits a VECM with the chosen rank using statsmodels, and prints beta normalised on the first variables, alpha with t-statistics and the half-lives of the equilibrium errors.`

Checks you must run

- The critical values: Engle–Granger (MacKinnon) for residuals, not Dickey–Fuller; Johansen for the right deterministic case
- The direction of the null: "no cointegration" in Engle–Granger; "rank $\leq r$" in the trace test
- That every series is $I(1)$ before testing, and that a high correlation of levels is not reported as cointegration
- The sign of the adjustment coefficients and the half-lives, in the right time unit
- Backtests: no look-ahead in pair selection, costs included, the number of pairs tested reported
- Every cited reference: it must exist; check the DOI

Key takeaways

- Cointegrated $I(1)$ series share common stochastic trends; a combination $\beta^\top \mathbf{y}_t$ is stationary
- Engle–Granger: OLS in levels, then ADF on the residuals with MacKinnon critical values; check with Phillips–Ouliaris
- ECM: changes respond to the lagged equilibrium error; γ gives the speed and the half-life
- VECM: $\Pi = \alpha\beta^\top$; Granger representation: cointegration $\Leftrightarrow$ error correction; Johansen tests choose r
- Cointegration helps forecast relations (spreads), not levels; applications must be judged out of sample and after costs

Key formulas

Quantity	Formula
Cointegration	$\mathbf{y}_t \sim I(1), \quad \beta^\top \mathbf{y}_t \sim I(0)$
Engle–Granger	$y_t = a + bx_t + u_t; \quad \Delta\hat{u}_t = \gamma\hat{u}_{t-\mathbf{1}} + \sum_j \delta_j \Delta\hat{u}_{t-j} + e_t, \quad H_\mathbf{0}\colon \gamma = 0$
ECM	$\Delta y_t = c + \gamma\,\hat{u}_{t-\mathbf{1}} + \delta_\mathbf{0}\Delta x_t + \cdots + \varepsilon_t, \quad h_{\mathbf{1/2}} = \ln 0.5/\ln(1+\gamma)$
VECM	$\Delta\mathbf{y}_t = \alpha\beta^\top \mathbf{y}_{t-\mathbf{1}} + \sum_{i=\mathbf{1}}^{p-\mathbf{1}} \Gamma_i \Delta\mathbf{y}_{t-i} + \mathbf{u}_t, \quad \Pi = \sum_i A_i - I, \quad \Gamma_i = -\sum_{j>i} A_j$
Johansen	$\lambda_{\mathbf{trace}}(r) = -T\sum_{i>r}\ln(1-\hat{\lambda}_i), \quad \lambda_{\max}(r) = -T\ln(1-\hat{\lambda}_{r+\mathbf{1}})$
Equilibrium errors	$\mathbf{z}_t = \beta^\top \mathbf{y}_t\colon \quad \mathbf{z}_t = (I + \beta^\top\alpha)\mathbf{z}_{t-\mathbf{1}} + \ldots$

Self-assessment

- **Question**: Engle–Granger with two variables gives $\tau = -3.1$ for $T = 300$. Do you reject "no cointegration" at 5%?
  - **Answer**: no: $-3.1 > -3.36$; with the Dickey–Fuller value (-2.87) you would wrongly reject
- **Question**: an ECM gives $\hat{\gamma} = -0.1$ on monthly data. What is the half-life?
  - **Answer**: $\ln 0.5 / \ln 0.9 \approx 6.6$ months
- **Question**: four $I(1)$ variables have cointegration rank 3. How many common trends drive them?
  - **Answer**: $n - r = 1$
- Next: Chapter 8, long memory: series between $I(0)$ and $I(1)$, fractional differencing and ARFIMA

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